14.2 Solution: I have tried to follow the same procedure outlined in Problem 14.15, but the elliptic orbit makes the calculation impossible. Instead, I have consulted and copied Landau's approach to this problem here (See Landau and Lifshitz, *Classtical Theory of Fields*, Section 70).

For a particle in periodic motion, the lowest nonvanishing moment is the dipole moment. To this end, we can use the result from Problem 9.9 (a) to obtain the power radiated in the n-th harmonic, with $\mathbf{p}_n(t) = \mathbf{p_n} e^{-in\omega_0 t}$, where ω_0 is the fundamental frequency. Notice that, by Problem 9.1, the $-n$ harmonic will also contribute to this frequency, we therefore will have that, in Gauss unit ($\epsilon_0 \to (4\pi)^{-1}$), the power radiated is

$$P_n = 2 \cdot \frac{2}{3c^3}|\ddot{\mathbf{p}}_n(t)|^2 = \frac{4(n\omega_0)^4}{3c^3}|\mathbf{p}_n|^2.$$

Here, $\mathbf{p}_n = e\mathbf{r}_n$, where e is the charge of the particle in periodic motion and

$$\mathbf{r}_n = \frac{1}{T}\int_0^T \mathbf{r}(t)e^{in\omega_0 t}dt = \frac{\omega_0}{2\pi}\int_0^{2\pi/\omega_0} \mathbf{r}(t)e^{in\omega_0 t}dt,$$

with $T = 2\pi/\omega_0$ as the period of the motion.

(a) To apply the above result to the elliptic orbit, we need to obtain the Fourier transform of the particle's trajectory. However, given the parameterization of the orbit, it is still not very easy. However, notice that

$$\mathbf{v}_n(t) = \frac{d\mathbf{r}_n(t)}{dt} = \frac{d}{dt}\left(\mathbf{r}_n e^{-in\omega_0 t}\right) = -in\omega_0 \mathbf{r}_n e^{-in\omega_0 t} = \mathbf{v}_n e^{-in\omega_0 t},$$

or equivalently

$$\mathbf{r}_n = \frac{i}{n\omega_0}\mathbf{v}_n.$$

If we can find $\mathbf{v}_n$, then we will be able to find $\mathbf{r}_n$.

This turns out to be easier. We can calculate $\mathbf{v}_n$ as the Fourier transform of the velocity,

$$\begin{aligned}
\mathbf{v}_n &= \frac{\omega_0}{2\pi}\int_0^{2\pi/\omega_0} e^{in\omega_0 t}\dot{\mathbf{r}}(t)dt \\
&= \frac{\omega_0}{2\pi}\int_0^{2\pi} e^{in\omega_0 t}d\mathbf{r}(t) \\
&= \frac{\omega_0}{2\pi}\int_0^{2\pi} e^{in(u-\epsilon\sin u)}\begin{pmatrix} -a\sin u \\ a\sqrt{1-\epsilon^2}\cos u\end{pmatrix} du.
\end{aligned}$$

Then,

$$\mathbf{r}_n = \frac{i}{2n\pi}\int_0^{2\pi} e^{in(u-\epsilon\sin u)}\begin{pmatrix} -a\sin u \\ a\sqrt{1-\epsilon^2}\cos u\end{pmatrix} du.$$

We can use the identity

$$e^{ix\cos\phi} = \sum_{m=-\infty}^{\infty} i^m J_m(x)e^{im\phi}$$

to perform the integrals. Since

$$e^{in(u-\epsilon\sin u)} = e^{inu}\sum_{m=-\infty}^{\infty}(-i)^m J_m(n\epsilon)e^{-im(u+\pi/2)},$$

for the x-component, we have

$$\begin{aligned}
r_{nx} &= -\frac{ia}{2n\pi}\int_0^{2\pi} e^{in(u-\epsilon\sin u)}\sin u du \\
&= -\frac{a}{4n\pi}\int_0^{2\pi} e^{inu}\sum_{m=-\infty}^{\infty}(-i)^m J_m(n\epsilon)e^{-im(u+\pi/2)}(e^{iu}-e^{-iu})du \\
&= -\frac{a}{2n}\left[J_{n+1}(n\epsilon)(-i)^{n+1}e^{-i(n+1)\pi/2} - J_{n-1}(n\epsilon)(-i)^{n-1}e^{-i(n-1)\pi/2}\right] \\
&= -\frac{a}{2n}(-)^{n+1}\left[J_{n+1}(n\epsilon) - J_{n-1}(n\epsilon)\right] \\
&= (-)^n\frac{a}{n}J_n'(n\epsilon).
\end{aligned}$$

Similarly,

$$\begin{aligned}
r_{ny} &= \frac{ia\sqrt{1-\epsilon^2}}{2n}\left[J_{n+1}(n\epsilon)(-i)^{n+1}e^{-i(n+1)\pi/2} + J_{n-1}(n\epsilon)(-i)^{n-1}e^{-i(n-1)\pi/2}\right] \\
&= (-)^{n+1}\frac{ia\sqrt{1-\epsilon^2}}{2n}\left[J_{n+1}(n\epsilon)+J_{n-1}(n\epsilon)\right] \\
&= (-)^{n+1}\frac{ia\sqrt{1-\epsilon^2}}{n\epsilon}J_n(n\epsilon).
\end{aligned}$$

Now, we have the power radiated in the k-th multiple of the fundamental frequency ω_0 as

$$P_k = \frac{4e^2}{3c^3}(k\omega_0)^4 a^2\left\{\frac{1}{k^2}\left[\left(J_k'(k\epsilon)\right)^2 + \frac{1-\epsilon^2}{\epsilon^2}J_k(k\epsilon)^2\right]\right\}.$$

(b) For circular orbit, $\epsilon \to 0$. In this limit,

$$J_k(k\epsilon) \to \left(\frac{k\epsilon}{2}\right)^k, \quad J_k'(k\epsilon) \to \frac{k}{2}\left(\frac{k\epsilon}{2}\right)^{k-1},$$

and only $k=1$ frequency will survive. Therefore,

$$P_1 = \frac{4e^2}{3c^3}\omega_0^4 a^2\left(\frac{1}{4}+\frac{1}{4}\right) = \frac{2e^2}{3c^3}\omega_0^4 a^2.$$

For hydrogen-like atoms, their radius for the n-th eigenstate is

$$a_n = \frac{n^2 a_0}{Z} = \frac{n^2\hbar^2}{Zme^2}.$$

The fundamental frequency can be determined classically, by Newton's second law,

$$m\omega_0^2 a_n = \frac{Ze^2}{a_n^2},$$

or

$$\omega_0^2 = \frac{Ze^2}{ma_n^3} = \frac{Ze^2}{m}\left(\frac{Zme^2}{n^2\hbar^2}\right)^3 = \frac{Z^4m^2e^8}{n^6\hbar^6}.$$

Therefore,

$$\omega_0 = \frac{Z^2me^4}{n^3\hbar^3}.$$

The reciprocal mean life time is then given by

$$\begin{aligned}
\frac{1}{\tau} &= \frac{P_1}{\hbar\omega_0} = \frac{2e^2}{3c^3}\frac{\omega_0^3 a_n^2}{\hbar} = \frac{2e^2}{3c^3}\frac{1}{\hbar}\left(\frac{Z^2me^4}{n^3\hbar^3}\right)^3\left(\frac{n^2a_0}{Z} = \frac{n^2\hbar^2}{Zme^2}\right)^2 \\
&= \frac{2Z^4me^10}{n^5\hbar^6c^3},
\end{aligned}$$

which agrees with Problem 14.21 (a).